

Anesthesia at Remote Locations MCQ – Chapter 37

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 37)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What chronic condition do patients at high altitudes (low barometric pressure) have?

- A. Chronic hypercapnia
- B. Chronic alkalosis
- C. Chronic hypoxia
- D. Chronic hyperoxia

Ans: C

Q2. What complications can chronic hypoxia at high altitude lead to?

- A. Only respiratory alkalosis
- B. Only polycythemia
- C. Only pulmonary hypertension
- D. Polycythemia, pulmonary hypertension, decreased pCO₂ and lower bicarbonate concentration

Ans: D

Q3. What happens to the potency of anesthetic gases at high altitudes?

- A. Decreases (proportional to partial pressure which decreases at high altitude)
- B. Remains same
- C. Becomes unpredictable
- D. Increases

Ans: A

Q4. At what altitude is the potency of nitrous oxide reduced to 50%?

- A. More than 3,300 meters
- B. 2,000 meters
- C. 5,000 meters
- D. 1,500 meters

Ans: A

Q5. At what altitude should nitrous oxide not be used?

- A. More than 3,000 meters
- B. More than 2,000 meters
- C. More than 4,000 meters
- D. More than 1,000 meters

Ans: B

Q6. What happens to the density of gases at high altitudes and how does it affect the rotameter?

- A. Density decreases, rotameter under reads the actual flow rate
- B. Density increases, rotameter over-reads
- C. Density decreases, rotameter over-reads
- D. No change

Ans: A

Q7. By how much can the rotameter error be at high altitudes?

- A. Up to 1 liter/minute
- B. Up to 4 liters/minute
- C. Up to 2 liters/minute
- D. Up to 6 liters/minute

Ans: B

Q8. At altered barometric pressure, do all vaporizers deliver fixed partial pressure of vapor? A. Only sevoflurane vaporizer B. All except desflurane deliver fixed partial pressure C. Yes, all vaporizers D. None of them	Ans: B
Q9. Which vaporizer needs upward adjustment at high altitude? A. Sevoflurane B. Desflurane C. Isoflurane D. Halothane	Ans: B
Q10. Why do Venturi type devices deliver higher oxygen at high altitude? A. Due to increased humidity B. Due to higher flow rates C. Because of decreased density and resistance D. Due to increased pressure	Ans: C
Q11. Why is blood loss during surgery more at high altitude? A. Due to coagulopathy B. Due to high venous pressure and blood volume C. Due to polycythemia only D. Due to hypothermia	Ans: B
Q12. Why are chances of postoperative hypoxemia high at high altitude? A. Due to low ambient partial pressure of oxygen B. Due to metabolic alkalosis C. Due to anemia D. Due to hyperventilation	Ans: A
Q13. What is the incidence of postspinal headache at high altitude under regional anesthesia? A. Very rare B. High (due to increased CSF pressure) C. Low D. Same as sea level	Ans: B
Q14. In hyperbaric chamber, what can happen to tissues with nitrous oxide? A. No effect B. Tissues become dehydrated C. Tissues can become supersaturated causing significant diffusion hypoxia D. Tissues become acidotic	Ans: C

Q15. Why should nitrous oxide not be used in helium atmosphere? A. It causes explosion B. It reacts with helium C. The effect of supersaturation is more evident in helium atmosphere D. It becomes toxic	Ans: C
Q16. Why does the rotameter falsely give high values in hyperbaric chamber? A. Due to humidity B. Due to temperature changes C. Due to low pressure D. Due to increased gas density	Ans: D
Q17. What should the endotracheal tube cuff be filled with in hyperbaric chamber? A. Helium B. Water or saline C. Air D. Nitrous oxide	Ans: B
Q18. What happens to the vaporizer output in hyperbaric chamber? A. Remains same B. Becomes unpredictable C. Becomes less D. Increases	Ans: C
Q19. Is regional anesthesia safe in hyperbaric chamber? A. Yes, regional anesthesia is safe B. No, it is contraindicated C. Only epidural is safe D. Only spinal is safe	Ans: A
Q20. What does anesthesia outside operating theater mean? A. Only emergency procedures B. Only dental procedures C. Field anesthesia only D. All procedures done outside the operation theater (radiology, MRI, CT scan, gastroscopy, cardiology, neurology etc.)	Ans: D
Q21. What are the challenges of procedure rooms outside OT? A. Only staffing issues B. Small and crowded, making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult C. Only equipment issues D. Only space issues	Ans: B

Q22. What equipment-related challenge exists in nonoperating room anesthesia? A. Only monitor issues B. Too much equipment C. Equipment is too modern D. Nonavailability of adequate equipment, monitors, technical support and help in case of emergency	Ans: D
Q23. In MRI suites, why does the anesthetist stand far away from the patient? A. Due to radiation B. Due to noise C. Due to temperature D. Due to the magnetic field (MRI suites require special non-ferrous equipment)	Ans: D
Q24. Why do medical proceduralists sometimes view anesthesia demand as obstructive? A. Unlike surgeons, many medical proceduralists do not understand the intricacies and importance of anesthesia B. Due to space constraints C. Due to scheduling conflicts D. Due to cost	Ans: A
Q25. How do patients undergoing medical procedures compare to surgical patients in terms of health status? A. They are the same B. They are always well-prepared C. They are healthier D. They are often even sicker than surgical patients and seldom evaluated and medically optimized preoperatively	Ans: D
Q26. What is the risk associated with radiology procedures requiring anesthesia? A. Risk of bleeding B. Risk of exposure to radiation C. Risk of infection D. Risk of hypothermia	Ans: B
Q27. What complication can occur in radiology procedures? A. Contrast-related anaphylaxis B. Only bleeding C. Only infection D. Only pain	Ans: A
Q28. Why may patients for nonoperating room procedures not be adequately fasting? A. They refuse to fast B. Fasting is not required C. They are always admitted D. Cases often come as unbooked, disturbing the schedule, and patients may not be adequately fasting	Ans: D

Q29. What is the full fasting recommendation for nonoperating room anesthesia? A. 8 hours for everything B. 4 hours for all C. 6 hours for light food, 8 hours for full meal, and 2 hours for water D. No fasting required	Ans: C
Q30. Why is inadequate sedation a concern in nonoperating room procedures? A. It causes awareness B. It can lead to conversion to general anesthesia, so full fasting is mandatory C. It causes pain only D. It is never a concern	Ans: B
Q31. What monitoring is highly desirable for spontaneously breathing patients in nonoperating room anesthesia? A. Only ECG B. Capnography (nasal capnography for spontaneously breathing patients) - can pick up life-threatening complications very early C. Only pulse oximetry D. Only blood pressure	Ans: B
Q32. What anesthesia technique is most often used for nonoperating room procedures? A. Either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl B. Spinal anesthesia C. Only local anesthesia D. General anesthesia with intubation	Ans: A
Q33. In India, what is used instead of Remifentanyl for TIVA in nonoperating room procedures? A. Sufentanyl B. Alfentanyl C. Morphine D. Fentanyl (due to nonavailability of remifentanyl)	Ans: D
Q34. What must all equipment in MRI suites be made of? A. Stainless steel B. Copper C. Aluminum (MRI compatible, nonferrous) D. Plastic only	Ans: C
Q35. Why is nonoperating room anesthesia considered very challenging? A. Due to long procedures only B. Due to patient refusal C. Due to small crowded space, inadequate equipment, inadequate preoperative evaluation, and inadequate knowledge of anesthesia intricacies by proceduralists D. Due to complex surgeries	Ans: C

Q36. Is nonoperating room anesthesia preferred by most anesthetists? A. Yes, most preferred B. Only preferred by junior anesthetists C. No, it is the least preferred by majority of anesthetists D. Equally preferred	Ans: C
Q37. What is the compensation for anesthetists working outside OT compared to surgical cases? A. No compensation B. Same compensation C. Almost always undercompensated financially compared to surgical cases D. Better compensation	Ans: C
Q38. Why is pCO₂ decreased at high altitude? A. Due to increased CO₂ production B. Due to renal compensation C. Due to decreased metabolism D. Due to hyperventilation (compensatory response to chronic hypoxia)	Ans: D
Q39. Why is bicarbonate concentration lower at high altitude? A. Due to dehydration B. Due to metabolic acidosis C. As a compensatory mechanism to low pCO₂ (renal compensation for respiratory alkalosis) D. Due to increased renal excretion of CO₂	Ans: C
Q40. At what altitude is it recommended to not use nitrous oxide? A. More than 4,000 meters B. More than 2,000 meters C. More than 1,000 meters D. More than 500 meters	Ans: B
Q41. What is the effect of high altitude on pulmonary circulation? A. Pulmonary hypotension B. Pulmonary hypertension C. Pulmonary vasodilation D. No effect	Ans: B
Q42. What happens to all vaporizers except desflurane at altered barometric pressure? A. They deliver fixed partial pressure of vapor so settings need not be changed B. They malfunction C. They deliver higher concentration D. They need recalibration	Ans: A

Q43. What is the fasting duration for water before nonoperating room anesthesia? A. 6 hours B. 1 hour C. 4 hours D. 2 hours	Ans: D
Q44. What is the fasting duration for a full meal before nonoperating room anesthesia? A. 8 hours B. 10 hours C. 6 hours D. 4 hours	Ans: A
Q45. What is the fasting duration for light food before nonoperating room anesthesia? A. 2 hours B. 6 hours C. 8 hours D. 4 hours	Ans: B
Q46. Can capnography detect life-threatening complications early in nonoperating room anesthesia? A. Yes, capnography can pick up life-threatening complications very early B. Only for cardiac complications C. Only in intubated patients D. No, it has limited use	Ans: A
Q47. Why must oxygen cylinders in MRI suites be made of aluminum? A. Aluminum stores more oxygen B. They must be MRI compatible (nonferrous) - ferrous materials are attracted by the magnetic field C. Aluminum is lighter D. Aluminum is cheaper	Ans: B
Q48. What type of polycythemia develops at high altitude? A. Polycythemia vera B. Relative polycythemia C. Secondary polycythemia (due to chronic hypoxia stimulating erythropoietin) D. Primary polycythemia	Ans: C
Q49. What is the main reason nonoperating room anesthesia cases are challenging regarding scheduling? A. Too few cases B. Complex procedures C. Too many cases D. Majority of cases come as unbooked, disturbing the schedule	Ans: D

Q50. Are full fasting recommendations required for non-operating room anesthesia?

- A. No, shorter fasting is acceptable
- B. Only for pediatric patients
- C. Only for GA
- D. Yes, full fasting recommendations are required

Ans: D

Answer Key & Explanations

Q1. Answer: C — Chronic hypoxia

The patients at high altitudes (low barometric pressure) have chronic hypoxia which can lead to polycythemia, pulmonary hypertension, decreased pCO₂ and lower bicarbonate concentration.

Topic: High Altitude

Q2. Answer: D — Polycythemia, pulmonary hypertension, decreased pCO₂ and lower bicarbonate concentration

Chronic hypoxia can lead to polycythemia, pulmonary hypertension, decreased pCO₂ (due to hyperventilation) and lower bicarbonate concentration (as a compensatory mechanism to low pCO₂).

Topic: High Altitude

Q3. Answer: A — Decreases (proportional to partial pressure which decreases at high altitude)

Potency of anesthetic gases is proportional to their partial pressure. At high altitudes where barometric pressure is decreased, the potency also decreases.

Topic: High Altitude

Q4. Answer: A — More than 3,300 meters

The potency of nitrous oxide is reduced to 50% at altitudes >3,300 meters.

Topic: High Altitude

Q5. Answer: B — More than 2,000 meters

It is recommended that nitrous oxide should not be used at altitudes more than 2,000 meters.

Topic: High Altitude

Q6. Answer: A — Density decreases, rotameter under reads the actual flow rate

Density of gases decreases at high altitudes, therefore, rotameter under reads the actual flow rate and this error may be up to 4 liters/minute.

Topic: High Altitude

Q7. Answer: B — Up to 4 liters/minute

The rotameter under reads the actual flow rate and this error may be up to 4 liters/minute.

Topic: High Altitude

Q8. Answer: B — All except desflurane deliver fixed partial pressure

At altered barometric pressure, all vaporizers (except desflurane) deliver fixed partial pressure of vapor. Therefore, vaporizer settings need not be changed at low or high barometric pressure except for desflurane.

Topic: High Altitude

Q9. Answer: B — Desflurane

Vaporizer settings need not be changed at low or high barometric pressure except for desflurane, which needs an upward adjustment at high altitude.

Topic: High Altitude

Q10. Answer: C — Because of decreased density and resistance

Venturi type devices deliver higher oxygen (because of decreased density and resistance).

Topic: High Altitude

Q11. Answer: B — Due to high venous pressure and blood volume

Blood loss during surgery is more due to high venous pressure and blood volume.

Topic: High Altitude

Q12. Answer: A — Due to low ambient partial pressure of oxygen

In the postoperative period, chances of hypoxemia are high due to low ambient partial pressure of oxygen.

Topic: High Altitude

Q13. Answer: B — High (due to increased CSF pressure)

The incidence of postspinal headache is high due to increased CSF pressure.

Topic: High Altitude

Q14. Answer: C — Tissues can become supersaturated causing significant diffusion hypoxia

Tissues can become supersaturated with nitrous oxide causing significant diffusion hypoxia.

Topic: Hyperbaric Chamber

Q15. Answer: C — The effect of supersaturation is more evident in helium atmosphere

The effect of supersaturation is more evident in helium atmosphere, therefore nitrous oxide should not be used in helium atmosphere.

Topic: Hyperbaric Chamber

Q16. Answer: D — Due to increased gas density

Rotameter will falsely give high values due to increased gas density.

Topic: Hyperbaric Chamber

Q17. Answer: B — Water or saline

Endotracheal tube cuff should be filled with water or saline.

Topic: Hyperbaric Chamber

Q18. Answer: C — Becomes less

The vaporizer output becomes less.

Topic: Hyperbaric Chamber

Q19. Answer: A — Yes, regional anesthesia is safe

Regional anesthesia is safe.

Topic: Hyperbaric Chamber

Q20. Answer: D — All procedures done outside the operation theater (radiology, MRI, CT scan, gastroscopy, cardiology, neurology etc.)

Nonoperating room anesthesia means all procedures done outside the operation theater. For example, anesthesia services utilized in radiology (MRI, CT scan), gastroscopy suites, intervention cardiology and neurology, etc.

Topic: Outside OT

Q21. Answer: B — Small and crowded, making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult

Procedure rooms are often small and crowded making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult.

Topic: Outside OT

Q22. Answer: D — Nonavailability of adequate equipment, monitors, technical support and help in case of emergency

Nonavailability of adequate equipment, monitors, technical support and help in case of emergency.

Topic: Outside OT

Q23. Answer: D — Due to the magnetic field (MRI suites require special non-ferrous equipment)

Many times, like MRI suites, anesthetist stands far away from the patient.

Topic: Outside OT

Q24. Answer: A — Unlike surgeons, many medical proceduralists do not understand the intricacies and importance of anesthesia

Unlike surgeons, many of the medical proceduralists do not understand the intricacies and importance of anesthesia. Any demand by anesthetist is often considered as obstructive by medical specialist.

Topic: Outside OT

Q25. Answer: D — They are often even sicker than surgical patients and seldom evaluated and medically optimized preoperatively

Majority of the patients undergoing medical procedures are even sicker than surgical patients. In spite of that, they are seldom evaluated and medically optimized in preoperative period.

Topic: Outside OT

Q26. Answer: B — Risk of exposure to radiation

Risk of exposure to radiation.

Topic: Outside OT

Q27. Answer: A — Contrast-related anaphylaxis

Contrast-related anaphylaxis can occur in radiology procedures.

Topic: Outside OT

Q28. Answer: D — Cases often come as unbooked, disturbing the schedule, and patients may not be adequately fasting

Majority of the times, cases come as unbooked and so, disturbing the schedule. Patients may not be adequately fasting.

Topic: Outside OT

Q29. Answer: C — 6 hours for light food, 8 hours for full meal, and 2 hours for water

Full fasting recommendation, i.e. 6 hours for light food, 8 hours for a full meal, and 2 hours for water is mandatory.

Topic: Outside OT

Q30. Answer: B — It can lead to conversion to general anesthesia, so full fasting is mandatory

Inadequate sedation leading to conversion to general anesthesia is very common, therefore, full fasting recommendation is mandatory.

Topic: Outside OT

Q31. Answer: B — Capnography (nasal capnography for spontaneously breathing patients) - can pick up life-threatening complications very early

Other than pulse oximetry, ECG and BP, capnography (nasal capnography for spontaneously breathing patients) is highly desirable; capnography can pick up life-threatening complications very early.

Topic: Outside OT

Q32. Answer: A — Either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl

Most often either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl is utilized for nonoperating room procedures.

Topic: Outside OT

Q33. Answer: D — Fentanyl (due to nonavailability of remifentanyl)

In India fentanyl, due to nonavailability of remifentanyl, is utilized for nonoperating room procedures.

Topic: Outside OT

Q34. Answer: C — Aluminum (MRI compatible, nonferrous)

For MRI suites, everything including anesthesia machine, monitors, oxygen cylinders must be MRI compatible, i.e. nonferrous. The equipments used in MRI suites are made up of aluminum.

Topic: Outside OT

Q35. Answer: C — Due to small crowded space, inadequate equipment, inadequate preoperative evaluation, and inadequate knowledge of anesthesia intricacies by proceduralists

Small and crowded space, nonavailability of adequate equipment, monitors, technical support and help in case of emergency, inadequate preoperative evaluation of patients and inadequate knowledge of anesthesia intricacies by medical proceduralists make nonoperating room anesthesia very challenging.

Topic: Outside OT

Q36. Answer: C — No, it is the least preferred by majority of anesthesiologists

Nonoperating room anesthesia is very challenging and least preferred by majority of anesthesiologists.

Topic: Outside OT

Q37. Answer: C — Almost always undercompensated financially compared to surgical cases

In spite of the high risk and adverse working circumstances, anesthesiologists are almost always undercompensated financially as compared to surgical cases.

Topic: Outside OT

Q38. Answer: D — Due to hyperventilation (compensatory response to chronic hypoxia)

Decreased pCO₂ is due to hyperventilation which is a compensatory response to chronic hypoxia at high altitude.

Topic: High Altitude

Q39. Answer: C — As a compensatory mechanism to low pCO₂ (renal compensation for respiratory alkalosis)

Lower bicarbonate concentration is a compensatory mechanism to low pCO₂.

Topic: High Altitude

Q40. Answer: B — More than 2,000 meters

It is recommended that nitrous oxide should not be used at altitudes more than 2,000 meters.

Topic: High Altitude

Q41. Answer: B — Pulmonary hypertension

Chronic hypoxia at high altitude can lead to pulmonary hypertension.

Topic: High Altitude

Q42. Answer: A — They deliver fixed partial pressure of vapor so settings need not be changed

At altered barometric pressure, all vaporizers (except desflurane) deliver fixed partial pressure of vapor. Therefore, vaporizer settings need not be changed.

Topic: High Altitude

Q43. Answer: D — 2 hours

Full fasting recommendation: 6 hours for light food, 8 hours for a full meal, and 2 hours for water.

Topic: Outside OT

Q44. Answer: A — 8 hours

Full fasting recommendation: 8 hours for a full meal.

Topic: Outside OT

Q45. Answer: B — 6 hours

Full fasting recommendation: 6 hours for light food.

Topic: Outside OT

Q46. Answer: A — Yes, capnography can pick up life-threatening complications very early

Capnography can pick up life-threatening complications very early. It is highly desirable for nonoperating room anesthesia.

Topic: Outside OT

Q47. Answer: B — They must be MRI compatible (nonferrous) - ferrous materials are attracted by the magnetic field

For MRI suites, everything including oxygen cylinders must be MRI compatible, i.e. nonferrous. The equipments used in MRI suites are made up of aluminum.

Topic: Outside OT

Q48. Answer: C — Secondary polycythemia (due to chronic hypoxia stimulating erythropoietin)

Chronic hypoxia at high altitude leads to polycythemia as a compensatory mechanism to increase oxygen carrying capacity.

Topic: High Altitude

Q49. Answer: D — Majority of cases come as unbooked, disturbing the schedule

Majority of the times, cases come as unbooked and so, disturbing the schedule.

Topic: Outside OT

Q50. Answer: D — Yes, full fasting recommendations are required

Full fasting recommendations are required for non-operating room anesthesia.

Topic: Outside OT